

## Problem 1

1. On the Gisette dataset, train decision trees of maximum depth  $1, 2, \dots$  up to 12, for a total of 12 decision trees. Use the trained trees to predict the class labels on the training and test sets and obtain the training and test misclassification errors. Plot on the same graph the training and test misclassification errors vs tree depth as two separate curves. Report in a table the minimum test error and the tree depth for which the minimum was attained.

We do this in Python, and obtain a plot of the training and test misclassification errors vs tree depth in figure 1:

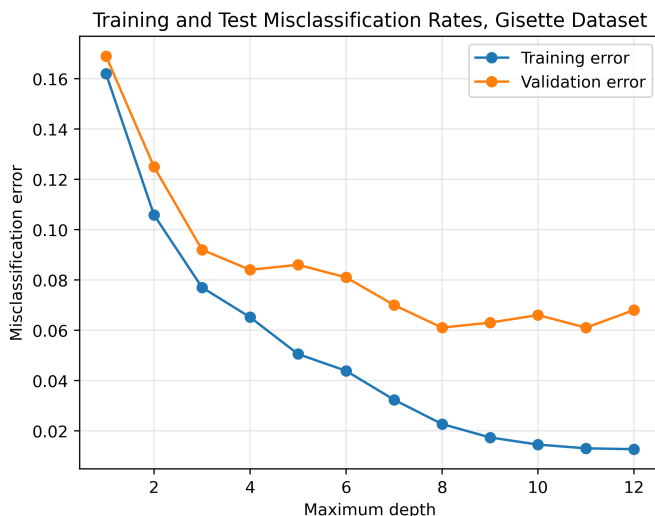


Figure 1: Misclassification Error vs Tree Depth, Training and Validation: Gisette dataset

The table freporting the minimum test error and the tree depth for which the minium was attained are displayed in table 1.

| Depth | Training Misclassification | Validation Misclassification |
|-------|----------------------------|------------------------------|
| 9     | 0.017500                   | 0.061000                     |

Table 1: Lowest misclassification error, Gisette Dataset

2. Repeat point 1.) on the satimage, madelon, hill-valley, and covtype datasets.

We can do this also, and obtain the four images shown in figure 2.

The table reporting the minimum test error and tree depth for the various datasets is shown in table 2

3. On the Gisette dataset, for each off  $k \in \{3, 10, 30, 100, 300\}$  train a random forest with  $k$  trees where the split attribute at each node is chosen from a random subset of  $71 \approx \sqrt{5000}$  features. Use the trained trees to predict the class labels on the training

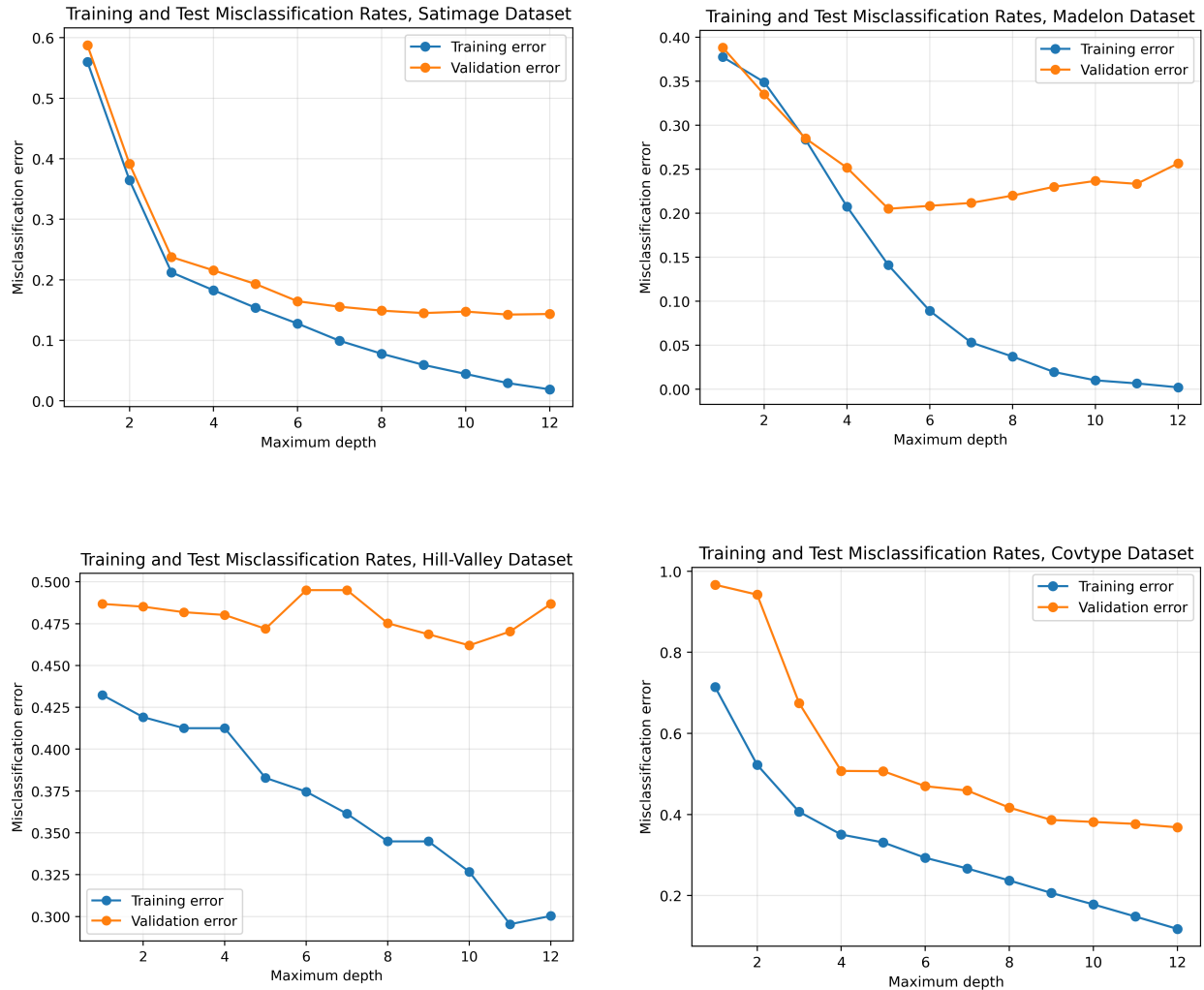


Figure 2: Training and Validation datasets misclassification error rates for four datasets

and test sets and obtain the training and test misclassification errors. Plot on the same graph the training and test errors vs. number of trees  $k$  as two separate curves. Report the training and test misclassification errors in a table.

We do this in Python, and obtain the plot of the training and test misclassification errors vs tree depth shown in figure 5:

The table reporting the training and test errors for various random forests are shown in table 3:

4. Repeat point 3.) on the satimage, madelon, and hill-valley datasets.

The images showing the training and test misclassification errors are displayed in figure 4 and we can create a table for this, shown in table 4.

5. Repeat point 3.) on the Gisette dataset where the split attribute at each node is chosen from a random subset of  $12 \approx \log_2(5000)$  features, and where the split attribute at

| Dataset     | Depth | Training Misclassification | Validation Misclassification |
|-------------|-------|----------------------------|------------------------------|
| Satimage    | 9     | 0.059076                   | 0.138000                     |
| Madelon     | 6     | 0.089000                   | 0.201667                     |
| Hill-Valley | 8     | 0.354785                   | 0.465347                     |
| Covtype     | 12    | 0.117659                   | 0.368692                     |

Table 2: Lowest misclassification error across datasets

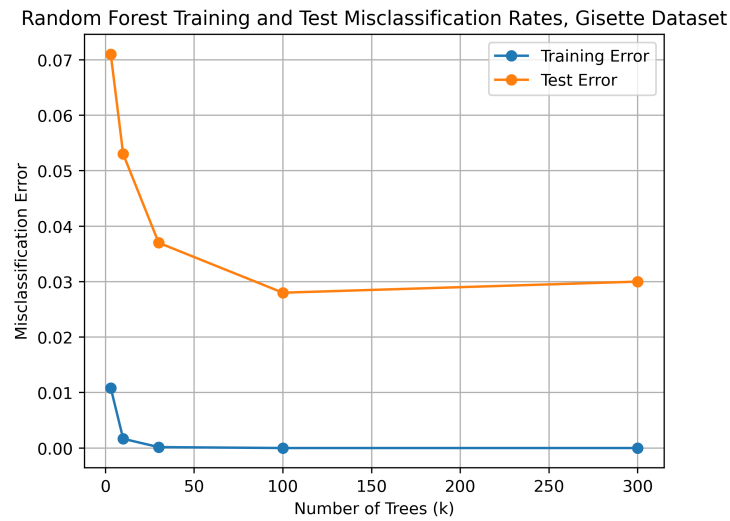


Figure 3: Misclassification Error vs Tree Depth, Training and Validation: Gisette dataset

each node is chosen from all 5000 features, plotting all four curves (2 train and 2 test misclassification errors) on the same graph.

Doing this gives us the following plot:

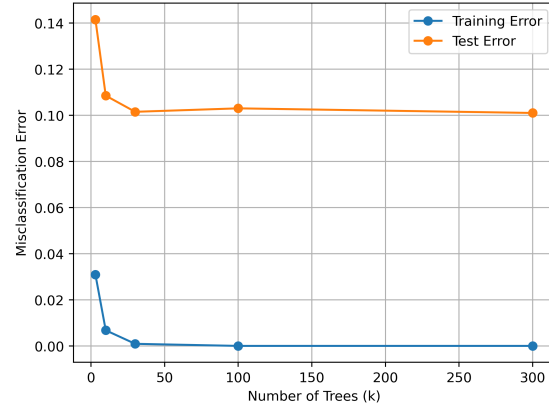
| Number of Trees | Training Misclassification Error | Validation Misclassification Error |
|-----------------|----------------------------------|------------------------------------|
| 3               | 0.011167                         | 0.064000                           |
| 10              | 0.001333                         | 0.047000                           |
| 30              | 0.000167                         | 0.033000                           |
| 100             | 0.000000                         | 0.029000                           |
| 300             | 0.000000                         | 0.034000                           |

Table 3: Misclassification errors, Random Forests, Gisette Dataset

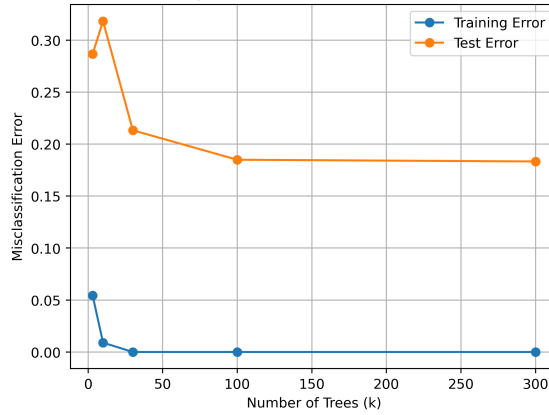
| Dataset     | Num. Trees | Training Misclass. Err. | Validation Misclass. Err. |
|-------------|------------|-------------------------|---------------------------|
| Satimage    | 3          | 0.027959                | 0.152000                  |
| Satimage    | 10         | 0.005637                | 0.114000                  |
| Satimage    | 30         | 0.000000                | 0.103500                  |
| Satimage    | 100        | 0.000000                | 0.101000                  |
| Satimage    | 300        | 0.000000                | 0.099500                  |
| Madelon     | 3          | 0.063000                | 0.325000                  |
| Madelon     | 10         | 0.010500                | 0.271667                  |
| Madelon     | 30         | 0.000000                | 0.228333                  |
| Madelon     | 100        | 0.000000                | 0.181667                  |
| Madelon     | 300        | 0.000000                | 0.175000                  |
| Hill-Valley | 3          | 0.080858                | 0.483498                  |
| Hill-Valley | 10         | 0.023102                | 0.460396                  |
| Hill-Valley | 30         | 0.000000                | 0.466997                  |
| Hill-Valley | 100        | 0.000000                | 0.453795                  |
| Hill-Valley | 300        | 0.000000                | 0.443894                  |

Table 4: Misclassification errors for Random Forests across datasets

Random Forest Training and Test Misclassification Rates, Satimage Dataset



Random Forest Training and Test Misclassification Rates, Madelon Dataset



Random Forest Training and Test Misclassification Rates, Hill-Valley Dataset

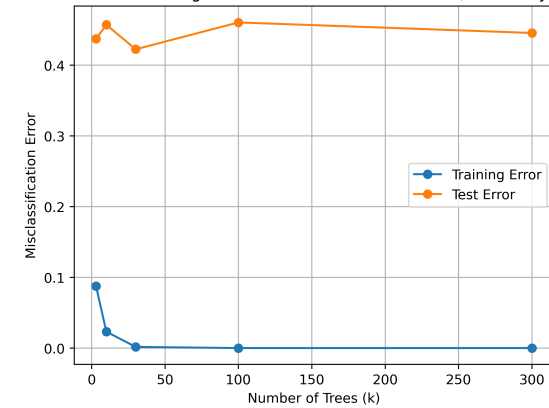


Figure 4: Plots Showing training and test misclassification errors vs number of trees.

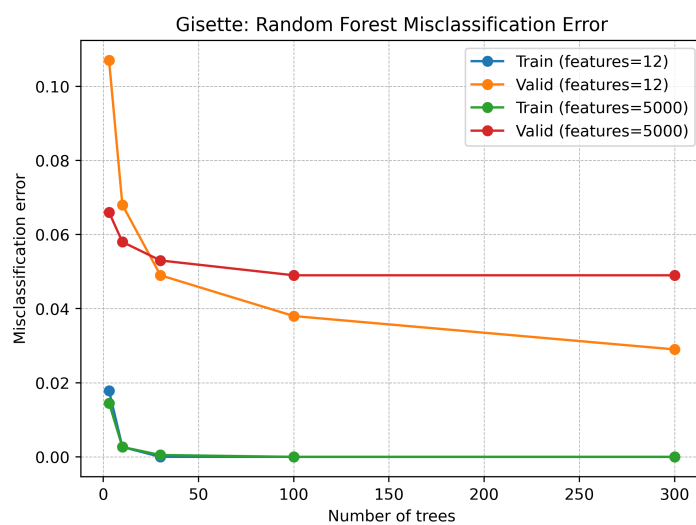


Figure 5: Misclassification Error vs Tree Depth, Training and Validation: Gisette dataset